CSE 4/546 Final Project: Behavioral Multi-Agent RL: Modeling Driver Experience and Psychology in F1 Race Strategy

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1 Short Summary

This project investigates the impact of human factors—specifically **experience, team adaptation, and psychological resilience**—on multi-agent Formula 1 race strategy. Following instructor feedback to ensure feasibility, this study will utilize an existing reinforcement learning (RL) environment rather than building one from scratch. We will model how non-rational performance variances, such as the initial "adaptation penalty" seen in elite transitions (e.g., Hamilton's team switch) or the high-variance consistency of 2025 rookies, affect strategic outcomes. By wrapping an existing racing simulator with behavioral modifiers, we aim to determine if a psychology-aware strategy can outperform standard baselines in high-pressure, multi-agent scenarios.

2 Objectives

- **Environment Adaptation:** Adapt an existing multi-agent racing environment (e.g., PettingZoo or Meta-Drive) to include a custom wrapper for behavioral state variables.
- **Behavioral Profile Implementation:** Define two primary agent modifiers:
  - **Experience Factor:** Controls the "noise" in action execution (low for veterans like Carlos Sainz, high for rookies).
  - **Adaptation/Psychology Factor:** Models a temporary performance dip for drivers in new teams or those prone to "tilt" after negative race events.
- **Strategic Evaluation:** Compare the performance of a standard MAPPO agent against one that explicitly observes driver "mental state" during a single-race strategic challenge.

3 Methodology

I will implement **Multi-Agent Proximal Policy Optimization (MAPPO)** as the primary algorithm, which aligns with the course's advanced policy gradient topics. To model driver behavior within the existing environment, I will introduce a **Driver State Vector** into the agent's observations:

- **Stochastic Execution:** The agent's chosen action will be perturbed by a noise function scaled by the driver's "Experience Level."

- **Recovery Mechanism:** Following a negative reward (e.g., an excursion or being overtaken), a "Psychological State" variable will degrade the agent's efficiency, recovering over a set number of laps.

This builds on the foundational DQN and Actor-Critic concepts from earlier assignments while providing the "novel idea" required for the final project.

4 Environment

As recommended by the TA, the project will use the **PettingZoo** library, extending a multi-agent racing wrapper. This avoids the overhead of building telemetry engines and allows focus on the strategic logic. The environment will simulate a discrete race session with variables for tire wear and pit-stop decision windows.

5 Evaluation

- **Qualitative:** Visualization of "Adaptation Curves" showing how a transferred elite agent's performance stabilizes as the "Team Familiarity" variable increases.
- **Quantitative:** We will measure the **Strategy Resilience Score**—defined as the ability to maintain a podium finish despite psychological setbacks—and compare lap-time consistency across Rookie vs. Veteran profiles.

6 Project Tracking

GitHub Project Board Link: `https://github.com/ub-cse546-s26/final-project-team-33`

7 References

1. Sutton, R. S., & Barto, A. G. (2019). *Reinforcement Learning: An Introduction.* MIT Press. http://incompleteideas.net/book/the-book-2nd.html
2. Fieni, G., et al. (2026). "Learning-based Multi-agent Race Strategies in Formula 1." ArXiv:2602.23056v1. https://arxiv.org/abs/2602.23056
3. Terry, J. K., et al. "PettingZoo: Gym for Multi-Agent Reinforcement Learning." https://pettingzoo.farama.org/